Bio-Silicon Labs

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Bio-Silicon Valley

BSLV Exam 2026 – Official Prospectus

Bio-Silicon Labs Vision (BSLV) Exam

Organized by Bio-Silicon Labs & DRY eWorks
(An Autonomous Research & Innovation Organization under Bio-Silicon Valley Initiative)

05 November 2025

Inviting Online Applications for Bio-Silicon Labs Vision (BSLV) Exam – 2026

The **Bio-Silicon Labs Vision (BSLV) Exam** is an International **Bioinformatics** Competitive examination for pursuing research in biotechnology to **NGS, AI-driven genomics, BigData handling, Quantum algorithms, Molecular simulation, XR, and advanced workflows**. **Apply** here : **www.biosiliconlabs.com/BSLV**

Important Dates, Fee & Application Procedure

Online Registration & Submission of Application Form	***05 November – 07 January***
Last Date of Fee Payment (Card / Net Banking)	***31 Jan 2026 (up to 11:50 PM UTC)***
Date of Examination (BSLV 2026)	***01 November 2026 (Sunday)***
Duration of Examination	***180 minutes (03 hours)***
Exam Timing	***10AM - 13 PM others & U.S 17 - 20 (UTC)***
Mode of Examination	***CLI-Based Bioinformatics Test*** *(Remote BSLV-Machine via Terminal)*
Syllabus	***Given in Information Bulletin Below***

Fee payable for BSLV– 2026:

Examination	Developing Countries	Developed Countries
BSLV	**₹1000 / EWS- ₹500**	**$100 / EWS- $50**

Sci. Abhinandan Yadav
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Founder Bio-Silicon Labs

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About the BSLV Exam

The Bio-Silicon Labs Vision (BSLV == Bio-Silicon Valley) Exam is a pioneering bioinformatics-specific competitive exam designed to identify and empower real-world talent on an international and global level. It is a unique initiative to:

- Fully reward top scorer with crowd funded amount of this exam for direct **#PhD/ #Scientist** Fellowship.
- Evaluate candidates through CLI-based real-time secure testing.
- Offer direct access to internships, R&D fellowships, and startup incubation.

Exam Details

Name: BSLV Exam 2026

Organizers: Bio-Silicon Labs, DRY eWorks & Vision@EnterprizeNX-01

ELIGIBILITY CRITERIA

Academic Qualification: Bachelors (B.Sc/ B.E./ B.Tech./ M.B.B.S.), Masters (M.Sc./ M.Tech./ M.VSc./ M.Pharm./ Integrated M.Sc./M.Tech.) and PhD, PostDoc in any discipline of Biotechnology, Life-Sciences or any other allied areas of **Bioinformatics** looking for direct **#PhD/ #Scientist** positions. Those who are yet to appear in the final semester examination or awaiting result of final semester examination are also eligible to apply.

Marks: Minimum 60% of marks for all aspirants (or equivalent grades).

Application Fee:
- Developing Countries (eg. India): ₹1000 (Standard), ₹500 (Weaker Sections))
- Developed Countries (eg. US, Europe: $100 (Standard), $50 (Weaker Sections))

Fellowships: for Direct PhD / Scientist position

- ~*1000 applicants from Developing Countries(**₹10,00,000** *Fellowship*)
- ~*350 applicants from Developed Countries(**$35,000** *Fellowship*)

Mode: CLI-based Online Exam

Top 100 Candidates will be called for the Interview Round at Bio-Silicon Labs.

Topper: Receives 100% of applicant pool fee as annual salary (e.g., 10,00,000 for *1000 developing country applicants and $35000 for developed country)

Next Top 10: Offered full-time R&D Internship

Next Top 50: Offered bootcamps, and courses

Exam Format

Practice Phase – System Setup & Terminal access Challenge

ssh candidateid@biosiliconvalley
BSLV-Machine: Virtual Machine Installation and testing for actual exam.

VM locks to full screen; only one-time ID login. Timed challenges (5–7 seconds per question). Application-based coding + reasoning - **human touch required*. If you know it, you make it. No questions repeated, No going back.

Phase 1 – Interactive CLI-Based Exam

Real-world questions on NGS, AI, Genomics & allied fields, with responses stored and shared as soon as the exam gets over, a copy stored in VM for authenticity.

PATTERN OF EXAMINATION

Type	Questions	Time / Q	Total Time
Ultra-short	400	5 sec	33 min 20 sec
Short MCQ	70	20 sec	23 min 20 sec
Medium reasoning	20	1 min	20 min
Long analytical	10	5 min	50 min
Active test time	500	—	2 h 6 min 40 sec

Breaks: 10-15 minutes after every 100 questions (4 breaks = 55 min).
Total Exam Time including breaks: **3hrs**.

Block-wise Structure

Block	Range	U/S/M/L	Block Time	Break	Cumulative Time
1	1–100	80/14/4/2	25:20	10:00	35:20
2	101–200	80/14/4/2	25:20	10:00	1:10:40
3	201–300	80/14/4/2	25:20	10:00	1:46:00
4	301–400	80/14/4/2	25:20	10:00	2:21:20
5	401–500	80/14/4/2	25:20	+15min(Review)	3:00:00 (END)

Post-Exam Feedback
Candidates receive a score sheet with score, percentile, and hash ID.

Phase 2 – *Top 100 Candidates will be called for the Interview Round at Bio-Silicon Labs. *Final selected Fellows(3-5) can choose* a 3-year **PhD** program or 1-year **Scientist** position**.** Other **Top 60 Candidates** receive full-time R&D Internship, Offered bootcamps, and courses.

Benefits

For **Top 60-70 Candidates**. *Final selected Fellows(3-5) can choose* a 3-year **PhD** program or 1-year **Scientist**, ***receiving the entire collected exam fee as a reward***. Next Top 10: Offered full-time R&D Internship, Next Top 50: Offered bootcamps, and courses.

Region	Top Performer Reward
Developing Countries	**₹10,00,000**
Developed Countries	**$35,000**

- Entry into the Bio-Silicon Valley innovation ecosystem
- Work with real-time data, Quantum -AI models, and biotech systems
- Our current work on Space Biotechnology, which is an interesting one and you can join us in that too.
- Direct collaboration opportunities with NGS and precision medicine pipeline dev work i.e for PRS.

Apply Now

Website: www.biosiliconlabs.com/BSLV

Contact: abhinandan@biosiliconlabs.com or +91 8802273055

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BSLV Exam Syllabus (Detailed Guide)

1. Core Bioinformatics with NGS

➔ **Sequence Analysis:** DNA, RNA, and Protein sequence alignment (global vs local, multiple sequence alignment).
➔ **File Formats:** FASTA, FASTQ, SAM/BAM, GFF, VCF - structure, usage, and parsing.
➔ **Reference Genome & Indexing:** Bowtie2, BWA, STAR - how reference indexing works.
➔ **NGS Pipeline Concepts:**

- Quality control (FastQC, MultiQC)
- Adapter trimming (Trimmomatic, Cutadapt)
- Mapping and alignment
- Variant calling (GATK, FreeBayes, bcftools)
- Expression quantification (featureCounts, HTSeq)
- Differential expression (DESeq2, edgeR)
- Annotations to gene functions
- Conclusive flowcharts

➔ **Transcriptomics:** RNA-seq pipeline overview; normalization (TPM, FPKM, RPKM).
➔ **Metagenomics:** Assembly vs binning; taxonomic profiling. Binning vs Mapping.
➔ **Epigenomics:** DNA methylation, ATAC-seq, and histone modification basics.
➔ **Data Integration:** Combining omics datasets (multi-omics approaches).

Points to Look For:

- Logic behind read alignment algorithms.
- Interpreting sequencing errors.
- Real-world applications (e.g., precision medicine, rare disease discovery).

2. Molecular Docking and Dynamic Simulations

➔ **Docking Fundamentals:**

- Types: Rigid, flexible, blind docking.
- Scoring functions and binding affinity.
- Docking software: AutoDock, Vina, Schrödinger Glide.

➔ **Ligand and Receptor Preparation:**

- Energy minimization.
- Water molecule role and removal.
- File formats (PDB, MOL2, PDBQT).

➔ **Molecular Dynamics (MD):**
- Force fields (CHARMM, AMBER, GROMOS).
- Solvation and ionization.
- Temperature and pressure control.
- Equilibration and production runs.

➔ **Post-simulation Analysis:**
- RMSD, RMSF, radius of gyration, hydrogen bonding.
- Free energy calculation (MM-PBSA/MM-GBSA).

➔ **Visualization Tools:** PyMOL, Chimera, VMD, UCSF ChimeraX.

Points to Look For:

- Understand energy landscapes.
- Interpretation of binding affinity and MD plots.
- Quantum docking vs. classical docking concepts.

3. Programming Knowledge

➔ **Languages:** Python, R, Shell scripting (focus on practical bioinformatics use cases).
➔ **Key Python Libraries:** Biopython, pandas, numpy, matplotlib, scikit-learn, TensorFlow, PyTorch.
➔ **Key R Packages:** DESeq2, ggplot2, dplyr, edgeR.
➔ **Command-Line Tools:** grep, awk, sed, cut, sort, uniq.
➔ **File I/O Operations:** Parsing FASTA, GFF, and VCF files.
➔ **Automation:** Snakemake, Nextflow, and workflow pipelines.

Points to Look For:

- Code understanding.
- Handling large biological datasets efficiently. File handling, ETL ways.
- Implementing algorithms for real problems (e.g., motif finding, clustering).

4. Database Development

➔ **Database Basics:** SQL vs NoSQL; Graph Dbs.
➔ **Schema Design:** Entity-relationship modeling for biological data.
➔ **Biological Databases:** GenBank, UniProt, PDB, Ensembl, GEO, KEGG, Reactome.
➔ **APIs and Integration:** Accessing data programmatically (NCBI Entrez, EBI APIs).
➔ **Web Development:** Flask/Django-based bioinformatics dashboards.
➔ **Data Curation:** Cleaning and annotating large-scale biological datasets.

Points to Look For:

- How biological databases are structured and maintained.
- Real-time data retrieval and storage methods.
- Efficient indexing and query optimization.

5. Hardware Details Required for Working

➔ **Basic Hardware Setup:** CPU, GPU, RAM requirements for NGS pipelines.
➔ **High-Performance Computing (HPC):** Clusters, SLURM job scheduling.
➔ **Parallel Computing:** MPI, CUDA, multithreading in genomics.
➔ **Cloud Platforms:** AWS, GCP, Azure for Bioinformatics workloads.
➔ **Data Storage and Backup:** SSDs vs HDDs, RAID systems, networked storage.

➔ **Edge & Quantum Devices:** Bioinformatics on embedded systems or quantum simulators.

Points to Look For:

- How computational resources affect bioinformatics workflows.
- Cost-effective system setups for research labs.
- Concept of quantum hardware for bio-data analysis.

6. Concepts Behind Software (e.g., BLAST, Seq to Structure Prediction, Live graphs …)

➔ **BLAST Algorithm:** Word seeding, scoring matrices (BLOSUM, PAM), alignment extension.
➔ **Algorithm:** Local alignment principle and heuristic search. MSA (ClustralW, MegaN).
➔ **Hidden Markov Models (HMMER):** Protein domain search.
➔ **Sequence Assembly:** De Bruijn Graph, Overlap-Layout-Consensus (OLC).
➔ **Read Mapping:** Smith-Waterman, Needleman-Wunsch, Burrows-Wheeler Transform (BWT).
➔ **Structural Prediction Tools:** AlphaFold, RoseTTAFold, MODELLER basics.
➔ **Full Stack Development:** Data fetching and hosting for live interactive graphs.

Points to Look For:

- Logic behind algorithmic design, not just tool usage.
- Complexity trade-offs (speed vs accuracy).
- Understanding the evolution of bioinformatics algorithms.

7. Algorithms / Concept to Code

➔ **Basic Algorithms:**
- Sorting, searching, graph traversal.
- Dynamic programming (alignment algorithms).

➔ **Bioinformatics-Specific:**
- Sequence alignment, clustering, motif search, gene prediction.
- Assembly algorithms (Contigs, Scaffold - SPAdes, Velvet).

➔ **Optimization Techniques:**
- Gradient descent, simulated annealing, genetic algorithms.

➔ **Complexity Analysis:** Big O notation and efficiency optimization.

Points to Look For:

- Convert theory to working code.
- Algorithmic thinking for biological data problems.
- Writing basic scripts for test datasets.

8. ML, DL, AI, GenAI, Agentic AI

➔ **Machine Learning:** Regression, classification, clustering, PCA.
➔ **Deep Learning:** CNNs, RNNs, Transformers in genomics and proteomics.
➔ **Generative AI:** Text-to-protein models, structure prediction using LLMs.
➔ **Agentic AI:** Autonomous agents in bioinformatics pipelines (AI-driven workflow automation).
➔ **Reinforcement Learning:** Applications in drug design and molecule optimization.
➔ **Explainable AI (XAI):** Feature interpretation for biological models.

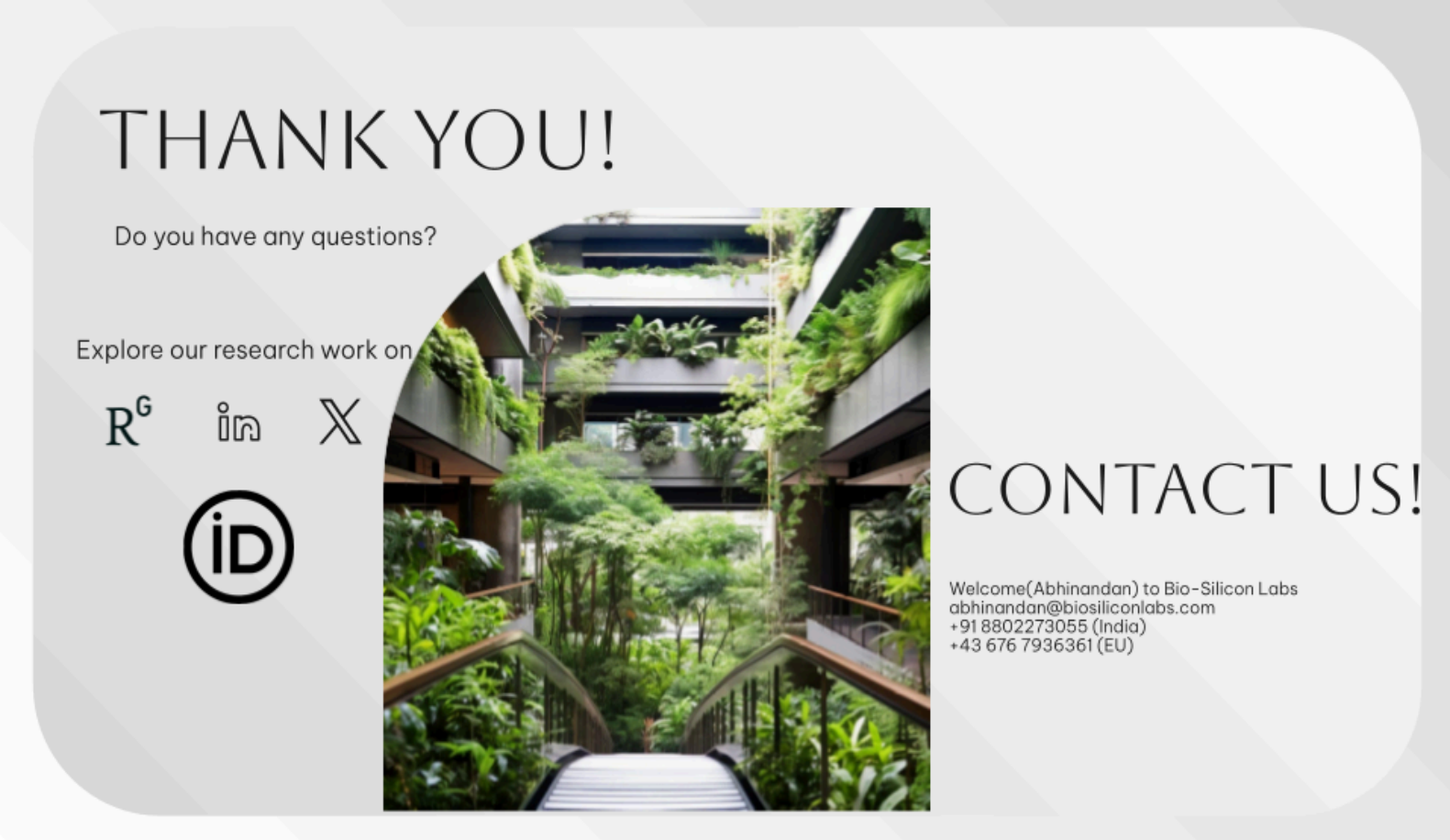
Points to Look For:

- Practical examples of ML/DL in NGS and drug discovery.
- Training vs inference trade-offs.
- Integration of AI models with biological databases.

9. Latest Bioinformatics Trends

➔ **Quantum Bioinformatics:** Quantum computing for NGS, structure prediction, and molecular docking.
➔ **Multi-Omics Integration:** Genomics, Transcriptomics, proteomics, metabolomics, epigenomics.
➔ **Single-Cell Analysis:** scRNA-seq pipelines, trajectory inference.
➔ **Synthetic Biology & Digital Twins:** In-silico simulation of biological systems.
➔ **Space Bioinformatics:** Radiation effects/Biomarkers evolution in DNA, bioreactor simulations for space.
➔ **XR in Bioinformatics:** AR/VR-based molecular visualization and education.
➔ **Bioinformatics for Personalized Medicine:** PRS, genotype-to-phenotype prediction.

Points to Look For:

- Awareness of cutting-edge technologies.
- Interdisciplinary integration mindset.
- Translational impact - from computation to real biology.

ALL the Best! @ Bio-Silicon Labs Vision